Lightweight Fuzzy-Driven Intrusion Detection for Consumer Life-Tech Applications

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***Abstract*—Consumer life-tech applications have significantly benefited from the rapid advancement of cutting-edge technologies, enabling the delivery of intelligent, cost-effective, reliable, and sustainable solutions. As consumer life-tech applications are being extensively embedded in innovative technologies, a key challenge is balancing security and resource efficiency in resource-constrained consumer devices. In this paper, we propose a lightweight fuzzy-driven intrusion detection framework to address these constraints by combining four key techniques: knowledge distillation, fuzzy logic integration, structured pruning, and quantization. We employ knowledge distillation to transfer decision-making capabilities from a large teacher model to a smaller student model. A fuzzy logic layer is further introduced to improve interpretability and robustness to uncertainties, while structured pruning and quantization are used to greatly reduce the model's computational and memory requirements. Our method achieves over 98% detection accuracy while greatly reducing model size and resource usage. This work offers a practical, interpretable, and high-performing intrusion detection solution for deployment in resource-constrained consumer life-tech ecosystems.**

***Index Terms*—Cybersecurity, fuzzy neural network, intrusion detection, IoMT, knowledge distillation.**

I. Introduction

THE RAPID advancement of innovative technologies has transformed consumer life-tech applications, delivering intelligent, cost-effective, autonomous, and sustainable solutions. Additionally, the Internet of Medical Things (IoMT) is a key consumer technology that has been widely used in the health-care sector. However, security in consumer health-care technology remain a significant challenge. For example, a hospital may use a smart health-care system to remotely monitor patients. IoMT devices, such as wearable electrocardiogram (ECG) monitors and glucose trackers, continuously transmit vital data to central systems. A cyberattack targeting these devices can compromise sensitive patient information and disrupt the monitoring process. In such a situation, health-care providers lose access to the critical data they need to make timely decisions, putting patients' lives at risk. To make matters worse, the attack depletes these devices' limited resources, leaving them inoperable and putting patients and caregivers in a difficult position. Unfortunately, existing security solutions often fail in these scenarios. Their high computational demands overwhelm resource-constrained devices, leading to significant performance issues. High false positive rates and overly complex algorithms further complicate the problem, making effective intrusion detection a considerable challenge. These challenges highlight the urgent need for a lightweight, efficient, and accurate intrusion detection mechanism that can safeguard consumer health-care technologies without straining their limited resources.

To address this need, this article proposes a novel approach that synergizes knowledge distillation, fuzzy logic integration, structured pruning, and quantization to create an efficient, robust, and lightweight intrusion detection system (IDS) for resource-constrained consumer technology. Transferring decision-making capabilities from large, high-performing models to smaller, resource-efficient ones ensures high detection accuracy with reduced computational and memory demands. The integration of fuzzy logic through a fuzzy layer handles uncertainties in network traffic by transforming input features into compact and informative representations. Structured pruning eliminates less significant neurons and connections to lower computational overheads, whereas quantization reduces the memory footprint by converting model weights into lower-bit representations. This comprehensive approach enhances performance on resource-constrained devices such as IoMT gadgets and smart appliances, and improves interpretability through fuzzy logic, fostering user trust.

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II. Related Work

Several studies have proposed different mitigation techniques for consumer technology. Zabeehullah et al. [1] developed an intelligent deep-learning-based model for detecting anomalies in IoMT networks, combining an autoencoder with deep learning models and achieving an improved accuracy. Narasimhan et al. [2] proposed an unsupervised deep learning approach for IDS, using an autoencoder for feature extraction and Gaussian mixture model for clustering into normal and attacks. Kalaria et al. [3] proposed the IoTPredictor for classifying attacks in smart devices, using hidden Markov models to detect cyberattacks and obtaining 88.6% accuracy. Alzubi et al. [4] proposed a deep learning approach for IoMT using a hybrid technique with convolutional neural network (CNN) and long short-term memory (LSTM) to identify attacks in consumer health-care networks. Rajesh et al. [5] proposed a threat detection model for IoT in health-care systems, selecting features with sigmoid cosine and pigeon-inspired techniques and using a hybrid ensemble model for attack classifications. Azimjonov and Kim [6] proposed

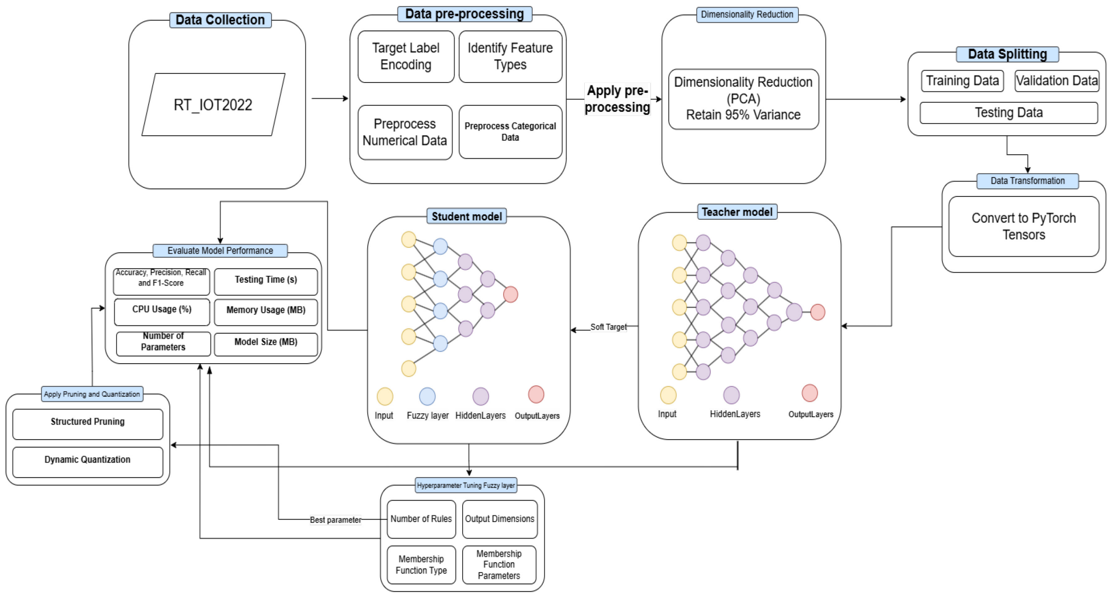

Fig. 1. Proposed architecture of fuzzy logic-knowledge distillation for IDS in resource-constrained consumer technology.

an intrusion detection model for detecting cyberattacks in IoT networks, using an importance coefficient for feature selection and stochastic gradient descent for IoT health-care. Singh et al. [7] proposed a SecureFlow detection framework using SNORT and support vector machine (SVM) to provide alerts of anomalies in IoT networks, demonstrating effectiveness in mitigating attacks. Some obstacles have been observed. **First,** consumer health-care technology's resource constraints pose challenges to ensuring effective mitigation of cyberattacks on consumer IoMT devices. **Second,** complexity and computational overheads lead to resource exhaustion, disrupting intrusion detection operations. **Third,** high false positive rates consume processing power and resources. **Fourth,** high data dimensionality affects detection performance, potentially causing overfitting and poor classifications. These issues highlight the need for robust intrusion detection mechanisms suitable for consumer health-care technologies. In the following, we aim to address these concerns.

III. Proposed Fuzzy Logic-Driven Knowledge Distillation for IDS in Consumer Technology

Fig. 1 illustrates the proposed fuzzy logic-driven knowledge distillation workflow for IDS in resource-constrained consumer technology. Data pre-processing and dimensionality reduction are performed using principal component analysis (PCA), retaining 95% of the variance. The data is then shuffled, split into training/testing sets, and converted to PyTorch tensors. A teacher-student framework is adopted, where the teacher trains a five-layer neural network from labeled data, while the simpler three-layer student model learns via knowledge distillation. A fuzzy student model further enhances decision-making and interpretability by incorporating a fuzzy layer. This layer transforms raw inputs into linguistic membership degrees (e.g., "low", "medium", "high") via a Gaussian membership function

$$\mu(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right), \quad (1)$$

where x is the input, c the center, and σ determines the width. Fuzzified data is processed via learnable fuzzy rules, enriching features with interpretable patterns for further classification. Further, the fuzzified data are processed by a set of fuzzy rules, which are represented as learnable weights that combine the membership degrees to meaningful outputs, which in turn are fed into successive layers of the student model for further processing through linear transformations and activation functions, yielding a binary classification decision. By embedding fuzzy logic, it may allow the student model to make sense of inputs in a more human-like reasoning manner, such as discussing events of probabilities, like the network packet being given classifications such as "medium" or "high" risk, in order to modify and improve both the interpretability of its decisions and its ability to adapt to complex input patterns. This allows the fuzzy student model to classify more knowledgeably and transparently, balancing accuracy, efficiency, and explainability. The loss function $\mathcal{L}_{\text{KD}}$ due to distillation is a combination of cross-entropy loss and KL-divergence loss between the teacher and the student models on their softened probabilities. This is represented as

$$\begin{aligned}\mathcal{L}_{\text{KD}} = \; & \alpha \cdot \text{CE}(y_s, y_{\text{true}}) \\ & + (1-\alpha) \cdot T^2 \cdot \text{KL}\left(\text{Softmax}\left(\frac{z_t}{T}\right), \text{Softmax}\left(\frac{z_s}{T}\right)\right), \quad (2)\end{aligned}$$

where α balances direct supervision and distillation guidance. T is the temperature for the softening probabilities. y_{true} is the ground truth label from dataset, y_s is the prediction output from student model. z_s and z_t are final layer output of student and teacher models, respectively.

The cross-entropy loss is defined as

$$\text{CE}(p, q) \; = -\sum_i p_i \log(q_i), \quad (3)$$

where i indexes the class labels, and p_i and q_i are the target and predicted probabilities for class i. The probability softening is controlled through a temperature parameter $T = 3$ and the relative weighting of the teaching guidance against

TABLE I
EXPERIMENTAL RESULTS FOR TEACHER MODEL, FUZZY STUDENT MODEL, AND PRUNED & QUANTIZED FUZZY STUDENT MODEL

Metric	Teacher Model	Fuzzy Student Model	Pruned & Quantized Fuzzy Student Model	Improvement (Pruned vs. Teacher)
Number of parameters	126,146	1,426	864	Significantly less parameters
Model size (MB)	0.485506	0.007909	0.008011	Smaller model footprint
Accuracy	98.334958	98.058804	98.176576	Slightly less accuracy
Precision	99.480519	99.583162	99.272661	Minor drop in precision
Recall	98.662268	98.251006	98.693903	Slight improvement in recall
F1 score	99.069704	98.912598	98.982436	Slight decrease in F1 score
Testing time (sec)	0.051194	0.011733	0.006552	Much faster inference
Average memory usage (MB)	353.05	348.95	346.25	Less memory consumption for training
Aggregate CPU Usage (%)	377.238741	366.820392	357.030568	Less CPU usage for training

direct supervision is given by the weighting factor $\alpha = 0.5$ to provide a balanced trade-off between learning from the teacher's soft labels and aligning with the true labels, yielding robust performance in our experiments. Advanced optimization techniques improve efficiency by applying L2-norm structured pruning, which removes 20% of the neurons from weight matrices to reduce model size and computational cost. Additionally, efficiency is enhanced through quantization, where floating-point weights are converted to 8-bit integers.

IV. EXPERIMENTAL EVALUATION

We assumed a deployment scenario where model training is carried out on an edge or cloud-enabled device, and the trained model is then transferred to a resource-constrained device for inference. We used the RT-IoT2022 dataset, introduced by Sharmila and Nagapadma in 2023 (Cybersecurity). Our training and testing were conducted on our lab machine with 12 logical core CPU to simulate the training and deployment phases under controlled conditions. We considered various performance metrics, including the number of parameters, model size, accuracy, precision, recall, F1 score, testing time, and CPU and memory usage during the training phase. Our results are reported in Table I, showing where improvements have been achieved. Fig. 2 illustrates the training time and total CPU utilization (across 12 logical cores) for all models under study. Our proposed model offers substantial improvements in terms of model size, memory usage, and inference speed, making it highly suitable for deployment on resource-constrained devices. While there are minor compromises in accuracy and precision, the overall detection performance remains strong and comparable to the teacher model. Overall, the proposed model achieves competitive detection performance while reducing the algorithm complexity and its computation time for resource-constrained consumer devices.

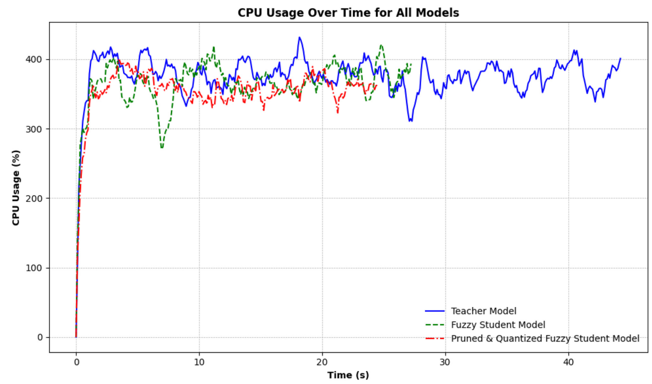

Fig. 2. Training time and CPU utilization for models under study.

V. CONCLUSION

In this paper, we presented a fuzzy-logic-driven knowledge distillation framework for intrusion detection in resource-limited consumer technologies. By combining knowledge distillation, fuzzy logic, structured pruning, and quantization, we achieved over 98% detection accuracy while greatly reducing model size and resource usage. Further, the fuzzy membership functions made the system more interpretable. Our proposed model is especially beneficial for consumer life-tech devices, which often have limited storage and processing power. In future work, we plan to explore adaptive fuzzy memberships, multi-class detection, and privacy-preserving methods to further strengthen security for a wide range of consumer applications.

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